ModelIt!

Teacher's Guide

How Social Media Hacks Your Brain

Modeling Dopamine, Feedback Loops, and Digital Addiction

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model of the brain's dopamine reward system to illustrate how hierarchical biological systems — from neurotransmitters to behavior to mental health — interact across scales.
Disciplinary Core Idea	**LS1.A: Structure and Function** — The nervous system's hierarchical organization — from neurotransmitter molecules to neural circuits to behavioral patterns — demonstrates how disruption at the molecular level (dopamine dysregulation) cascades to affect the whole organism's functioning and well-being.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from external stimulus (notification) to molecular response (dopamine release) to behavioral change (increased screen time) to psychological outcome (self-esteem change), demonstrating multi-level cause and effect.

CAST Testing Alignment

MULTIPLE CHOICE:

A student turns off all push notifications on their social media apps but still checks them manually every 15 minutes. According to your model, which component is most responsible for maintaining the compulsive behavior?

CONSTRUCTED RESPONSE:

Using your model, explain the complete causal chain from a single Instagram notification to a measurable decrease in self-esteem. Identify every component involved and whether each relationship is positive or negative.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Your brain evolved over millions of years to seek rewards that helped your ancestors survive — food, social bonding, mating opportunities. Every time you found something rewarding, your brain released dopamine to say 'do that again.' Social media has hijacked this ancient system, delivering hyper-concentrated social rewards at a speed and frequency your brain was never designed to handle.

The Dopamine Reward Pathway

Dopamine is released by neurons in the ventral tegmental area (VTA) and projects to the nucleus accumbens and prefrontal cortex. When you receive a like, comment, or new follower, this pathway activates — the same circuit triggered by food, sex, and drugs. The key insight is that dopamine doesn't just signal pleasure; it signals ANTICIPATION of reward. The buzz of a notification is often more powerful than the actual content, because your brain is responding to the POSSIBILITY of a reward.

Variable Reward Schedules

B.F. Skinner discovered in the 1950s that the most addictive reward pattern is variable ratio reinforcement — rewards that come at unpredictable intervals. Slot machines use this principle: you never know when the next win is coming, so you keep pulling. Social media does the same thing. You don't know which refresh will bring a flood of likes or a disappointing nothing. This unpredictability keeps you checking, because the next time MIGHT be the big payoff.

Social Comparison Theory

Psychologist Leon Festinger proposed in 1954 that humans have a fundamental drive to evaluate themselves by comparing to others. On social media, you're comparing your unfiltered daily life to everyone else's curated highlight reel — their best photos, achievements, and moments. Research shows this 'upward social comparison' consistently decreases self-esteem and life satisfaction, particularly in adolescents whose identity is still forming.

Neuroplasticity and Digital Habits

The brain physically changes in response to repeated experiences — neurons that fire together wire together. Heavy social media use strengthens the neural pathways associated with phone-checking behavior while weakening pathways for sustained attention, delayed gratification, and real-world social skills. Brain imaging studies show that adolescents who use social media heavily have measurable differences in their prefrontal cortex (decision-making) and amygdala (emotion processing) compared to light users.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify notification frequency, screen time, dopamine release, social comparison, and self-esteem level as the key components of the social media–brain interaction system.

E **Establish Relationships**

Students determine that notifications trigger dopamine release, dopamine drives screen time, screen time enables social comparison, and social comparison erodes self-esteem — forming a self-reinforcing positive feedback loop.

V **Visualize & Model**

Students build a computational model showing the positive feedback loop between social media design features and the brain's reward system, and the downstream effects on self-esteem.

E **Evaluate Outcomes**

Students run low-notification, high-notification, and gradual escalation scenarios to test their model and identify the tipping point where use becomes compulsive.

R **Revise & Extend**

Students add sleep quality, anxiety level, or algorithm intensity to explore more realistic models of social media's impact on adolescent well-being.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The average American teen spends 4.8 hours a day on social media. That's more time than sleeping, doing homework, or being with family. Why?"*

Watch For

Show the statistic on screen. Let it sink in. Ask: Does that number sound right for you? Higher? Lower?

Slide 2: Learning Objectives

3 min

Say: *"Today we're reverse-engineering the technology that's designed to hack your brain. Once you see the code, you can't unsee it."*

Watch For

Have students read objectives. Pre-teach 'dopamine' and 'variable reward schedule.' Ask who knows how slot machines work.

Slide 3: Big Question

3 min

Say: *"You've told yourself a hundred times: 'five more minutes.' An hour later, you're still scrolling. That's not weakness — that's engineering."*

Watch For

Quick-write: Students describe a time they couldn't stop scrolling even though they wanted to. Pair-share.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a model of what's happening inside your skull every time your phone buzzes."*

Watch For

Preview LEVER steps. Emphasize this is about understanding the SYSTEM, not shaming anyone's phone use.

Slide 5: Activity 1: Components

8 min

Say: *"What's the one thing the app CONTROLS that starts the whole chain? And what are the things that happen inside your brain and mind?"*

Watch For

Guide sorting: notification frequency is external (the platform controls it). Everything else is internal to the user. Discuss why this matters.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the chain: notification hits your phone. Then what? What fires in your brain? What do you DO? And how do you FEEL after?"*

Watch For

Help students build the complete causal loop. Identify the positive feedback loop and explain why it escalates.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's find out: what happens if you turn notifications OFF? And what happens if they're on all the time?"*

Watch For

Students predict outcomes, then run all three scenarios. Key discovery: the tipping point from casual to compulsive use.

Slide 8: Discoveries

5 min

Say: *"Social media isn't a product. YOU are the product. The app is the machine designed to harvest your attention."*

Watch For

Discuss the ethics of addictive design. Compare to tobacco industry knowingly designing addictive products. Is this different?

Slide 9: STEM Challenge

12 min

Say: *"Now that you've seen the code, write the antidote. Design an app that helps people break the loop."*

Watch For

Groups design digital wellness tools using model insights. Must address the dopamine loop AND social comparison. Present prototypes.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Notification Frequency
- **INTERNAL:** Screen Time, Dopamine Release, Social Comparison, Self-Esteem Level

Relationships

- **Notification Frequency to Dopamine Release:** POSITIVE (+) — More frequent notifications create more opportunities for the brain's reward system to activate. Each buzz or badge triggers an anticipatory dopamine response, even before the user sees the content.
- **Dopamine Release to Screen Time:** POSITIVE (+) — Higher dopamine release reinforces phone-checking behavior. The brain learns that opening the app produces a reward, driving increased screen time as the user seeks to replicate the pleasurable feeling.
- **Screen Time to Social Comparison:** POSITIVE (+) — More time spent scrolling through feeds means more exposure to curated images of other people's lives, achievements, and appearances, intensifying the social comparison process.
- **Social Comparison to Self-Esteem Level:** NEGATIVE (-) — Increased social comparison — especially upward comparison to idealized, filtered content — erodes self-esteem as individuals perceive themselves as less attractive, successful, or popular than their peers appear online.

Key Discoveries

- Social media platforms exploit the brain's dopamine reward system using the same variable reward schedule as slot machines — unpredictable likes and notifications are more addictive than predictable ones
- A positive feedback loop forms: notifications trigger dopamine, dopamine drives more screen time, more screen time means more social comparison, and lower self-esteem drives users to seek more online validation
- The tipping point from healthy to compulsive use is driven by notification frequency — once the brain is conditioned to expect constant rewards, it becomes difficult to disengage
- Self-esteem erosion is gradual and often invisible to the user, making the damage harder to recognize until it's deeply established in neural pathways

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Social media addiction isn't real — it's just a lack of willpower"	Social media platforms are engineered by teams of behavioral psychologists and neuroscientists to be as habit-forming as possible. The same brain circuits activated by social media are activated by gambling and substance use. Brain imaging studies show measurable changes in the prefrontal cortex and reward pathways of heavy social media users. Calling it 'willpower' ignores the neuroscience.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"I use social media because I want to — it's my choice"	While the initial download is a choice, the continued use is heavily influenced by design features specifically created to bypass conscious decision-making. Variable reward schedules, autoplay, infinite scroll, and notification timing are all optimized through A/B testing to maximize engagement. Your 'choice' is being made in an environment engineered to produce a specific outcome.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Only weak or lonely people get addicted to social media"	Research shows that social media's addictive mechanisms work on ALL brains — the dopamine reward system is universal. In fact, highly social people may be MORE susceptible because they have a stronger need for social feedback. Adolescent brains are particularly vulnerable because the prefrontal cortex (impulse control) isn't fully developed until age 25.